



Polystyrene nanoplastics exacerbate gentamicin-induced nephrotoxicity in adult rat by activating oxidative stress, inflammation and apoptosis pathways

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Received: 22 November 2024 / Accepted: 8 January 2025

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Abstract

Nanoplastics (NPs) and pharmaceutical residues are environmental pollutants that can bioaccumulate in the food chain, thus increasing the risk of harmful effects due to concomitant exposure in both humans and animals. However, the impact of such co-exposure on target tissue toxicity and mechanism remains unclear. In this study, we aimed to investigate the effect of combined exposure to widespread polystyrene NP (Ps-NP) contaminant and a widely used antibiotic, gentamicin (GEN) on nephrotoxicity in adult rat. A total of 40 male Wistar rats were assigned into four groups; control group (C), GEN exposed-group (100 mg/kg/day, *i.p.*), Ps-NPs exposed-group (2.5 mg/kg/day, orally) and co-exposure-group (100 mg Ps-NPs/kg/day orally + 2.6 mg GEN/kg/day *i.p.*) for 15 consecutive days. Kidneys were excised at sacrifice on day 15 for the examination of: (i) renal function parameters and renal oxidative stress (oxidant and antioxidant markers); (ii) mRNA expression of key regulators of oxidative stress and inflammation response (NF- κ B, TNF- α , IL-6 and NRF-2) and mitochondrial apoptosis pathway (Bax, Bcl-2, caspase-9 and caspase-3); (iii) histopathology. Levels of urea and creatinine in serum of NPs or/and GEN-treated rats were significantly increased with marked histopathological changes in the kidney. Additionally, the oxidative stress markers (MDA, PCO and NO) were increased with a decrease of antioxidant enzyme activities. Furthermore, Ps-NPs or/and GEN-treated rats exhibited increased expression of mRNA related to oxidative stress (NRF-2), inflammation (TNF- α , IL-6) and apoptosis (Bax, Bcl-2, caspase-9 and caspase-3). Our results revealed that co-exposure of adult rats to Ps-NPs and GEN had a synergic-adverse effect on the kidney function. This study highlights a new finding regarding the combined toxicity of NPs and pharmaceutical pollutants on kidney function, suggesting potential synergistic interaction between these substances.

Keywords Nanoplastics · Oxidative stress · Inflammation · Gentamicin · Nephrotoxicity

Introduction

Nowadays, plastics have received a significant worldwide interest and concern due to their pervasive use in both society and industry. This widespread reliance on plastic materials has led to an alarming increase in their release into the environment, resulting in serious contamination issues (Abbasi and Turner 2021). Many varieties of plastics exist, and they are frequently manufactured for one-time use. Polyethylene, polypropylene, polystyrene, polyvinyl chloride

(PVC), polyamide, and polyethylene terephthalate (PET) are among the most prevalent (Öztekin and Sponza 2023). The annual production of plastic has seen a significant rise, from 250 million tons in 2009 to 299 million tons in 2013, and further escalating to 335 million tons by 2016 (Otitoju et al. 2023). Microplastics (MPs) and nanoplastics (NPs) pose a potentially greater threat compared to macroplastics, as their smaller size allows them to penetrate biological membranes. MPs are defined as synthetic or fragmented plastic pieces measuring less than five mm in diameter (Stock et al. 2020), while NPs represent the colloidal component of plastic waste (Lambert and Wagner 2016; Garcia-Muñoz et al. 2023). Primary NPs are those produced through industrial manufacturing processes, while secondary NPs are the result of environmental degradation of larger plastic items (Dube and Okuthe

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2023). MPs and NPs, including both primary and secondary forms, are classified as emerging pollutants (Kumar et al. 2023) due to the lack of comprehensive understanding surrounding their potentially harmful structural characteristics and long-term impacts on living beings.

The widespread presence of plastic particles has generated significant concerns regarding human health (Ragusa et al. 2021; Yee et al. 2021). However, the effects of human exposure to these plastic particles remain mostly uncertain, as the bulk of research has focused on detecting their presence and assessing their effects on environmental systems, instead of exploring their direct effects on human health (Bradney et al. 2019; Xu et al. 2022). Researchers have identified packaged water bottles as a primary source of human exposure to NPs (with varying concentrations ranging from 11 to 6290 items and 0.4 to 11.4 µg/L), as well particle sizes of less than 5 µm found in the packaging materials, such as bottles and caps. Oral ingestion is the main route of exposure in the general population, as recent studies have detected significant amounts of nanoplastics in food items, potable water, and via the utilization of plastic materials that interact with food. MPs (particularly those in the sub-micron range, 0.1 to 1 µm) and NPs (under 100 nm) may penetrate cell membranes and surpass the intestinal barrier, potentially entering the bloodstream and migrating to other organs (Naidu et al. 2024). Health effects associated with the intake of MPs and NPs could stem from the plastic particles themselves as well as from their additives or the pollutants they adsorb (Pathak and Singh 2023). Owing to their large specific surface area and a variety of functional groups, NPs easily attract pollutants such as heavy metals, organic pollutants and drugs, potentially altering their bioavailability and toxicity (Amato-Lourenço et al. 2020).

Currently, data regarding the effects of NPs on mammalian systems are limited, with most research have especially focused on their chemical properties. Indeed, the harmful impacts of leachate from particles have been investigated across a range of organisms, including sea urchins (Rendell-Bhatti et al. 2021) and mussels (Gonçalves et al. 2023), while the behavioral and physical effects remain poorly understood.

Polystyrene, a highly produced plastic type, is also a primary constituent of plastic debris that pollutes the environment (Zheng et al. 2023). Due to their small size, PS-NPs can effortlessly penetrate biological membranes, resulting in their build-up within tissues and possibly inflicting physiological damage (Tang et al. 2022). Exposure to polystyrene NPs triggers reactive oxygen species production (ROS), which can result in endoplasmic reticulum stress and oxidative stress (Das 2023; Wu et al. 2024).

Due to their small size, high stability, extensive surface area, broad distribution, and the presence of polyfunctional groups, nanoplastics (NPs) are commonly employed to

evaluate biological effects in conjunction with other contaminants (Gao et al. 2023). Recent studies have revealed the presence of antibiotic residues in aquatic environments (Roy et al. 2021). Nanoplastics (NPs) can serve as carriers for various pollutants, including antibiotics. This function enhances the accumulation of antibiotics in organisms in a manner that depends on size, which may lead to considerable toxicity (Yilimulati et al. 2021). Organisms can absorb highly bioactive antibiotics along with NPs, thereby elevating the risk of combined exposure (Xu et al. 2023). Therefore, exploring the combined effects of NPs and antibiotics on organism development, as well as the underlying mechanisms, holds significant theoretical and practical importance.

Gentamicin (GEN), an antibiotic from the aminoglycoside class, is extensively utilized and has received approval in many countries as treatment for infections caused by aerobic gram-negative bacilli (Sojo-Dorado and Rodríguez-Baño 2017; Corona et al. 2023). Moreover, GEN is a nephrotoxic antibiotic associated with acute tubular necrosis, which raises significant concerns regarding its toxicity in clinical practice. After being absorbed by organisms, it is metabolized and then discharged into wastewater treatment plants and other water bodies, presenting risks to both individual health and the broader ecosystem. Prolonged treatment with these drugs (GEM) can lead to concerning side effects. Studies have shown that administering aminoglycosides to patients for more than seven days results in nephrotoxicity in around 30% of treated cases (Ali 2003). Moreover, GEN-induced nephrotoxicity is responsible for 10–15% of all cases of acute renal failure (Pedrañez et al. 2004). GEN has been shown to stimulate the production of superoxide anions, hydrogen peroxide and hydroxyl radicals following its accumulation in renal proximal tubular cells (Morales et al. 2010). Additionally, some researchers have demonstrated that GEN acts as an iron chelator, and the combination of iron and gentamicin effectively enhances the generation of free radicals (Yanagida et al. 2004). The mechanism underlying gentamicin nephrotoxicity is not completely clear.

To date, no published studies are available concerning the adverse effects of NPs and pharmaceutical drugs (e.g., gentamicin) resulting from their combined exposure on the kidney tissue of male rats. Herein, this is the first work that aims to evaluate the possible nephrotoxic effects of both PS-NPs and GEN administered to adult rats as well as their potential interactions on oxidative stress, inflammation and apoptosis pathways.

Materials and methods

Drugs and chemicals

Polystyrene nanoplastics (PS-NPs, Cat. No: 07307–15), measuring 50 nm in diameter, were purchased from

Polysciences Europe GmbH in Germany. These nanoparticles were supplied in an aqueous solution containing 2.5% solids (w/v), resulting in a particle concentration of 3.64×10^{11} particles/ml. Gentamicin Sulfate injection (2 ml: 20 mg/ml) was obtained from Medis pharmaceutical Co., Ltd. (Nabeul—Tunisia). All other chemicals, of analytical grade, were acquired from commercial suppliers.

Animals and experimental design

Adult Wistar rats (male; 12–14 weeks; weighing around 200 ± 20 g) were provided from the Central Pharmacy, SIPHAT-Tunis. The animals were housed in clean polypropylene cages and were provided with a controlled environment (12-h light/dark cycle, temperature of 22 ± 2 °C, relative humidity at 55%).

The rats had free access to food and water. The quarantine and acclimation period extended for 7 days, during which no signs of distress or notable fluctuations in body weight were detected. The animal maintenance protocol received approval from the Ethical Commission for the Welfare Protection of Experimental Animals at the Sciences Faculty, Sfax University—Tunisia. Additionally, all experimental procedures were performed to the National Institutes of Health guidelines for the Care and Use of Laboratory Animals (1986).

After the acclimation period, the rats were weighed and then randomly assigned to four groups, each consisting of 10 animals:

- (i) Rats in the control group received either normal saline intraperitoneally (i.p.) or distilled water by gavage for a duration of 15 days;
- (ii) Rats in the PS-NPs group were received 2.5 mg/kg/day solely by gavage for a duration of consecutive 15 days;
- (iii) Rats in the GEN group were administered 100 mg/kg/day of GEN via i.p. for a period of 15 consecutive days;
- (iv) Rats in the GEN + PS-NPs group were administered 100 mg/kg/day of gentamicin via intraperitoneal injection, followed by the administration of PS-NPs at a dose 2.5 mg/kg/day by gavage, for 15 days. The exposure dose and particle size of PS-NPs were selected based on their reported concentrations in aquatic environments and findings from recent toxicological studies (Lin et al. 2019; Zheng et al. 2023). The specific dosage for PS-NPs exposure was established through toxicity assessments conducted on cell cultures in previous experiments, as well as references from recent research highlighting the potential

nephrotoxic effects of PS-NPs (Amereh et al. 2020; Ahmed et al. 2022; Zhu et al. 2023).

On the 14th day, 24-h urine was collected from rats housed in metabolic cages and centrifuged (5 min; 3000 g; 4 °C). Then, the resulting supernatants were collected and stored (at -80 °C). At the end of the experimental period (15 days), all rats were sacrificed by cervical decapitation method to avoid stress conditions. Whole blood samples were collected and then centrifuged (10 min; $3000 \times g$; 4 °C). Serum samples were drawn and stored (-80 °C) to perform biochemical analysis. Kidney tissues were quickly excised and rinsed with NaCl solution (0.9%). After being gently blotted dry with filter paper, some portions of kidney were frozen immediately (-80 °C) for biochemical and molecular analysis. Other kidney sections were fixed in 10% neutral-formalin and then embedded in paraffin for histopathological analysis.

Animal behavior observation

The body weights of the rats were measured and recorded during the treatment period. Furthermore, the animals were carefully observed for any changes in their general behavior, physical appearance, morphology, and signs of distress or mortality, such as alterations in facial color, eye appearance, paw condition, fur quality, presence of bleeding, and characteristics of excreta.

Estimation of renal function markers

Biochemical analyses of serum and urine samples, such as blood urea nitrogen (BUN), urea (UR), uric acid (UA), and creatinine (CREA) levels, were conducted with a Vitalab Flexor E automated biochemistry analyzer (Element RC, USA) in the Biochemistry Department of CHU-Hedi Chaker in Sfax (Tunisia).

Determination of glomerular filtration rate (GFR)

The glomerular filtration rate (GFR) was estimated using the UV/P equation, where U represents the concentration of creatinine in the urine measured in mg/dL, V denotes the volume of urine produced per minute, and P represents the concentration (mg/dL) of plasma creatinine (Shahbaz et al. 2024).

Determination of fractional excretion of urea (FEurea)

The fractional excretion rate of urea (FEurea) (Carvounis et al. 2002) was determined as follows:

$$F_{\text{Eurea}} = \frac{\text{Uurea} \times \text{PCr}}{\text{Purea} \times \text{UCr}} \times 100$$

where U urea refers to the level of urea measured in the urine, P urea indicates the level of urea in the plasma, UCr represents the level of creatinine in the urine, and PCr denotes the creatinine level in the plasma.

Determination of renal oxidative stress markers

Assessment of lipid peroxidation levels

The assessment of lipid peroxidation in kidney tissue was performed through a calorimetric approach that quantifies thiobarbituric acid reactive substances (TBARS), expressed as malondialdehyde (MDA) concentrations, following the methodology established by Draper and Hadley (Draper and Hadley 1990). MDA values were expressed as nM of TBARS per milligram protein.

Assessment of nitric oxide production

Nitric oxide (NO) production in kidney tissue was measured using the Griess Method (Green et al. 1982). Nitrite concentrations were determined by referencing a sodium nitrite standard curve, with absorbance readings taken at 540 nm, and results expressed in micromoles per liter (μM).

Assessment of protein carbonyl (PCO) levels

Protein carbonyl (PCO) content was determined as described by (Reznick and Packer 1994). The results were reported as nanomoles of carbonyl per milligram of protein.

Assessment of advanced oxidation protein product (AOPP) levels

The kidney AOPP level was determined spectrophotometrically following the protocol of (Witko-Sarsat et al. 1996). The absorbance of each sample was measured at 340 nm using the extinction coefficient of $26,000 \text{ M}^{-1} \text{ cm}^{-1}$.

Assay of renal enzymatic antioxidant activities

The SOD activity was assessed following the method reported by (Asada et al. 1974). The results were expressed as units per milligram of protein (U/mg protein). The measurement of GPx activity was conducted following the method of (Flohé and Günzler 1984) and was quantified as μmoles of glutathione (GSH) oxidized/min/mg protein. The assessment of CAT activity was measured by hydrogen peroxide decomposition using (Aebi 1984) technique. During H_2O_2 breakdown, absorbance decreased at 240 nm for

1 min and the enzyme activity was measured as $\mu\text{mol H}_2\text{O}_2$ degraded /min/mg protein.

Determination of GSH and NPSH levels

Glutathione (GSH) levels in kidney tissue were performed using Ellman's method (Ellman 1959), as modified by (Jollow et al. 2008). A yellow color was developed when DTNB interact with sulfhydryl groups present in the compounds. The absorbance (412 nm) was taken and the specific total GSH values was expressed as nanomoles per milligram of protein. The assay of non-protein thiols (NPSH) was performed according to the method of Ellman (Ellman 1959). The NPSH level was quantified spectrophotometrically at 412 nm and the results expressed as nanomoles per milligram of protein.

Protein determination

Protein content in kidney samples was quantified using the Bradford method (Bradford 1976), with bovine serum albumin (BSA) used as the calibration standard.

RNA isolation and real-time quantitative PCR

mRNA quantification analysis was carried out using quantitative reverse transcription PCR (qRT-PCR) with SYBR Green PCR Master Mix, adhering to the standardized protocol described by (Chtourou et al. 2016). The total RNA of kidney was extracted using the ExScript™ RT-qPCR Sample Preparation Reagent (170–8898, Bio-Rad), following the manufacturer's guidelines. The obtained RNA was assessed by A260/A280 ratio to check its concentrations and purity using NanoPhotometer™ (Implen, GmbH). The purified RNA (4 μg) was subjected to cDNA synthesis by adding 500 ng of oligo-dT primer (Thermo Fisher Scientific Inc.) and incubated at 70 °C for 10 min followed by the addition of Superscript II reverse transcriptase (Invitrogen). Quantitative RT-PCR assay was performed using the obtained cDNA template and 2 $\times$ Power SYBR Green (Osaka, Japan) containing specific primers (Table 1). Following PCR amplification, the expression levels of each target gene were calculated by subtracting the β actin Ct (control and normalization) from each sample Ct using the delta-delta Ct ($\Delta\Delta\text{Ct}$) method.

Histopathological investigations

For histological analysis, kidney tissues were preserved in 10% neutral formaldehyde for 72 h. The specimens were then embedded in paraffin blocks. Thin sections, measuring 5 μm , were deparaffinized with xylene, rehydrated

Table 1 Sequence of all primer used in RT-qPCR experiment

Gene name	Primer sequences	
	Forward	Reverse
<i>B-Actin</i>	5'- GGAGATTACTGCCCTGGCTCCTA -3'	5'- GCCAAGGTGTCAGATGATGTCC-3'
<i>IL-6</i>	5'- TCCTACCCCAACTTCCAATGCTC -3'	5'- TTGGATGGTCTTGGTCCTTAGCC-3'
<i>TNF-α</i>	5'- AAATGGGCTCCCTCTCATCAGTTC -3'	5'- CTGCTTGGTGGTTTGCTACGAC -3'
<i>NF-KBP65</i>	5'- CAAGATCAATGGCTACACGGGA -3'	5'- CGGACCGCATTCAAGTCATAGTC -3'
<i>Bax</i>	5'-AAACTGGTGCTCAAGGCC-3'	5'-GGGTCCCGAAGTAGGAAAGG-3'
<i>Bcl-2</i>	5'-GCTACGAGTGGGATACTGG-3'	5'-GTGTGCAGATGCCGGTTCA-3'
<i>NrF2</i>	5'-CAGCACATCCAGACAGACACCA-3'	5'-CGTATTAAGACACTGTAAC TCGGG AATGG-3'
<i>Caspase-9</i>	5'-AGATGCTGTCCCATACCAGG-3'	5'-CAGGAACCGCTCTTCTTGTC-3'

through a sequence of graded alcohol solutions, and subsequently stained with hematoxylin–eosin (H&E), Masson's trichrome, Sirius Red, and toluidine blue dyes. The specimens were examined at a magnification of 100×; 200×; 400× using a light microscope (Olympus BX 52, Tokyo, Japan). Three sections from each group of animals were subjected to microscopic analysis.

Statistical analysis

The acquired data were analyzed using GraphPad Prism 9.0 (GraphPad Software, San Diego, CA, USA) and presented as mean ± SEM. Statistical comparisons among the different groups were conducted using one-way analysis of variance (ANOVA), followed by Tukey's post hoc test. A *p*-value of less than 0.05 was considered statistically significant.

Results

Behavioral clinical signs, body, and organ weight changes

During the entire experimental period, all rats did not show any clinical signs or deaths. However, a significant decrease in the body weight was recorded in rats receiving both Ps-NPs and GEN when compared to the other two groups (Fig. 1). Changes in the body weight were not observed in the rats treated with Ps-NPs and GEN groups as compared with controls. There were no significant differences in the relative kidney weight in all treated groups when compared to the control group.

Effects of Ps-NPs and/or GEN exposure on renal function markers

Table 2 exhibits data concerning kidney specific biomarkers (creatinine, urea and uric acid levels). The

results showed a significant increase in creatinine, urea and uric acid levels of treated rats with GEN, Ps-NPs and Ps-NPs + GEN in plasma and decreased in urine when compared with controls. No significant differences were observed in the urea concentration of Ps-NPs group compared to the controls. Indeed, blood urea nitrogen levels (BUN) was evaluated. Levels of BUN were found to be significantly elevated in all treated rats when compared to controls. GEN-treated group exhibited a constellation of renal injury which was accompanied by an increase in 24-h urine volume.

Effects of Ps-NPs and/or GEN exposure on GFR level and FE urea

Compared to controls, rats treated with GEN and Ps-NPs or their combination showed disorders in renal function which are manifested by a significant increase in GFR level, considered as an indicator of glomerular dysfunction (Fig. 2). The fractional excretion of urea (FE) (%) was significantly lowered in groups treated by GEN and Ps-NPs or their combination when compared to controls; whereas no significant differences were observed in GFR level and FE urea between treated groups (GEN, Ps-NPs or their combination).

Effects of Ps-NPs or/and GEN exposure on the oxidative markers in the kidneys

The results of oxidative stress indicators in kidneys are shown in Fig. 3A–D. Compared to controls, the levels of MDA, PCO, AOPP and NO were increased significantly in the GEN, Ps-NPs and Ps-NPs + GEN groups. These values were higher in the co-treated rats (Ps-NPs + GEN) than those in the Ps-NPs group. No significant difference in the oxidative markers among treated rats with GEN, Ps-NPs or Ps-NPs + GEN was observed.

Figure 4 clearly illustrates the disruption in antioxidant enzymatic and non-enzymatic activity in kidney

Fig. 1 Effects of Ps-NPs or/ and GEN on body and relative kidney weights of adult rats. **A** Body weight changes in rats; **B** kidney coefficient changes in rats. Values are means $\pm$ SEM ($n = 10$). a: significant when compared to controls at $p < 0.05$; b: significant when compared to GEN group at $p < 0.05$; c: significant when compared to Ps-NPs group at $p < 0.05$

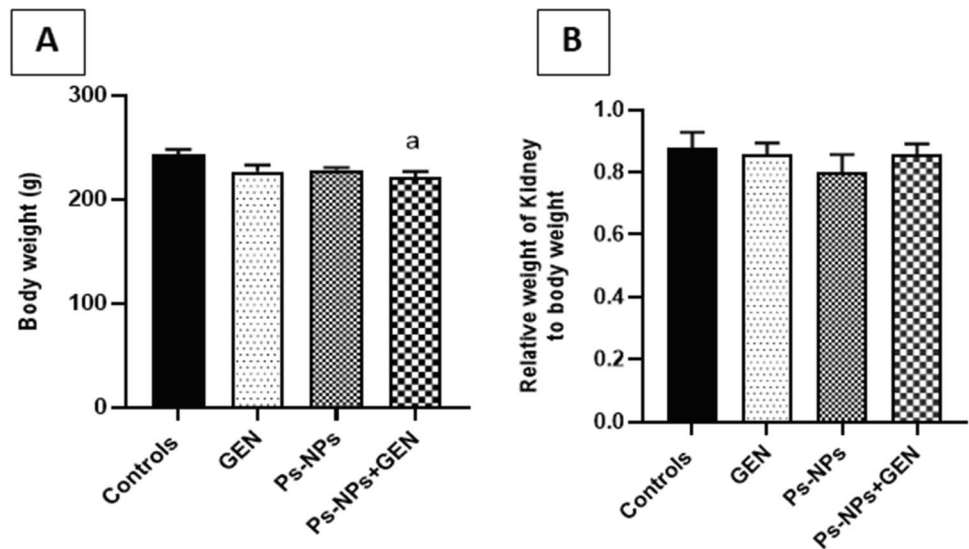
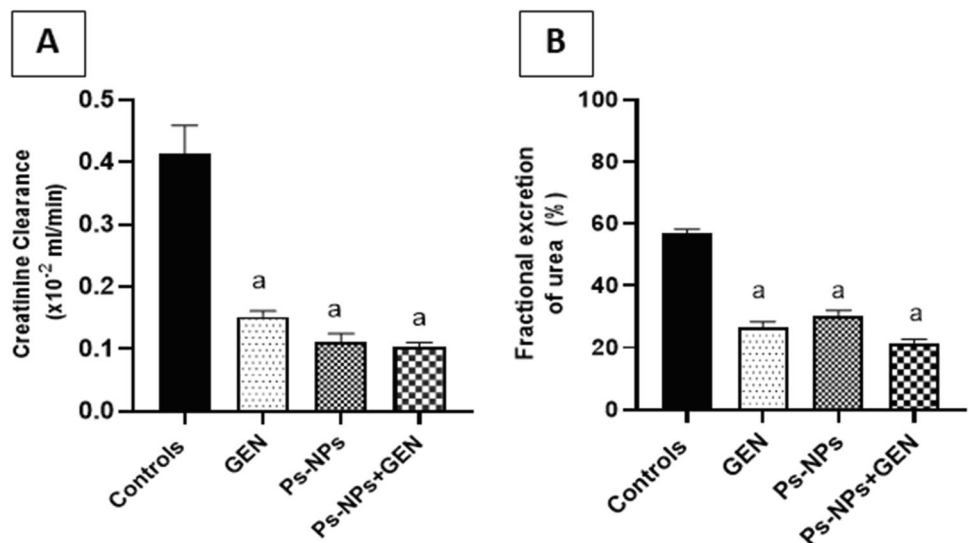


Table 2 Effects of Ps-NPs or/and GEN on renal biomarkers (creatinine, urea and uric acid) in plasma and urine of adult rats

Plasma concentration					Urine concentration		
Groups ($n = 10$)	Creatinine (mg/dl)	Urea (mg/dl)	Uric acid (mg/dl)	BUN (mg/dl)	Creatinine mg/dl	Urea (mg/dl)	Uric acid (mg/dl)
Control	0.58 ± 0.04	38.64 ± 2.01	0.59 ± 0.13	18.05 ± 0.94	52.28 ± 6.11	520.57 ± 15.92	20.75 ± 2.36
GEN	$1.14 \pm 0.11^{a,b}$	47.41 ± 3.81^a	0.96 ± 0.32^a	21.58 ± 1.56^a	24.96 ± 2.51^a	386.61 ± 18.41^a	15.84 ± 2.98^a
Ps-NPS	1.01 ± 0.05^a	38.56 ± 2.12	1.09 ± 0.58^a	17.55 ± 0.67^a	21.75 ± 5.43^a	$325.12 \pm 16.36^{a,b}$	$11.75 \pm 3.92^{a,b}$
Ps-NPS + GEN	$1.17 \pm 0.08^{a,c}$	42.01 ± 5.91^a	$1.29 \pm 0.31^{a,c}$	19.71 ± 0.31^a	20.82 ± 2.46^a	$398.25 \pm 11.35^{a,b,c}$	$10.08 \pm 2.48^{a,b}$

Values as means $\pm$ SEM ($n = 10$). a: significant when compared to control at $p < 0.05$; b: significant when compared to GEN alone $p < 0.05$; c: significant when compared to Ps-NPs alone at $p < 0.05$

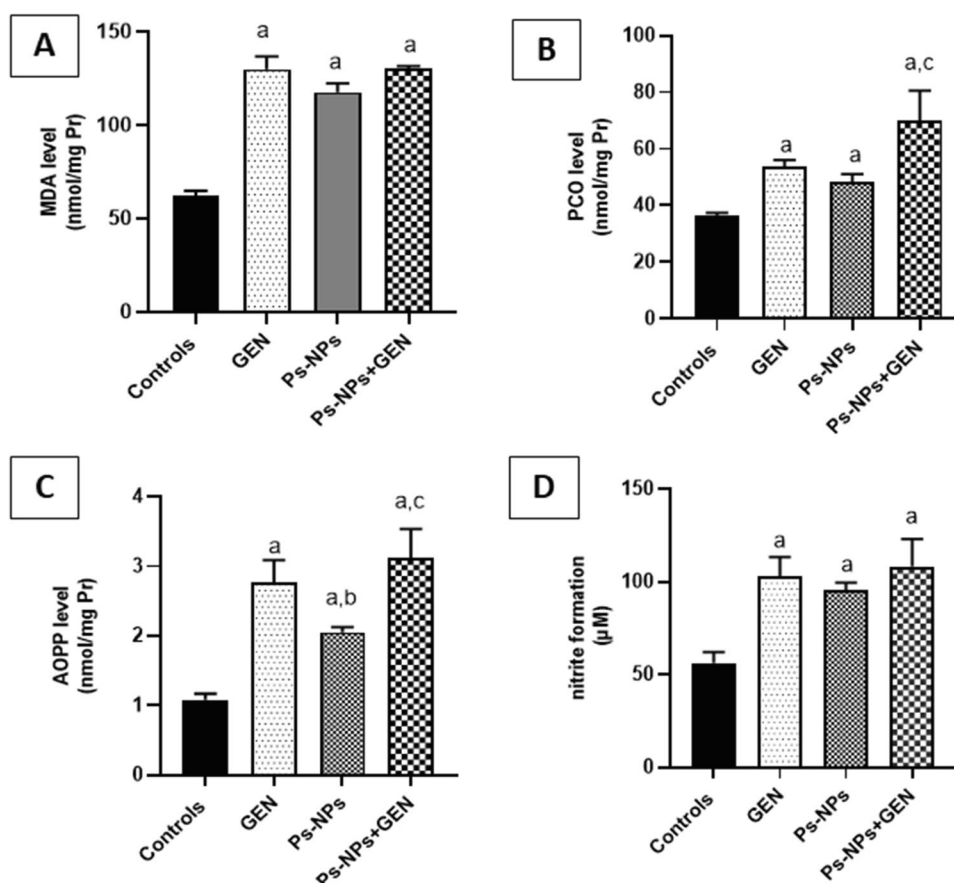
Fig. 2 Effects of Ps-NPs or/ and GEN on creatinine clearance (GEF) and fractional excretion of urea (FE). Values are means $\pm$ SEM ($n = 10$). **A** significant when compared to controls at $p < 0.05$; **B** significant when compared to GEN group at $p < 0.05$; **C** significant when compared to Ps-NPs group at $p < 0.05$



tissues. Activities of GPx, CAT and SOD were substantially decreased in Ps-NPs and Ps-NPs + GEN groups when compared to the controls. A significant increase of kidney CAT activity was observed in received Ps-NPs in the comparison to GEN group. No exposure-related

changes were observed in GPx activity between treated groups (GEN, Ps-NPs, Ps-NPs + GEN). Furthermore, administration of GEN and Ps-NPs or their combination to adult rats significantly decreased NPSH and GSH levels when compared to the controls. Group.

Fig. 3 Effects of Ps-NPs or/and GEN on renal oxidative stress markers (A), (B), (C), (D). Values are means $\pm$ SEM ($n=10$). **A** Significant when compared to controls at $p<0.05$; **B** significant when compared to GEN group at $p<0.05$; **C** significant when compared to Ps-NPs group at $p<0.05$



Effects of Ps-NPs or/and GEN exposure on gene expression related to oxidative stress and inflammation

Data related to the mRNA levels of pro-inflammatory cytokines such as TNF- α , IL-6, and transcription factors such as NF- κ B and Nrf2 in the kidney tissue are presented in Fig. 5. TNF- α expression was increased significantly in GEN and PS-NPs + GEN treated groups while no significant change was recorded in Ps-NPs group. Conversely, oral administration of Ps-NPs to adult rats significantly increased IL-6 expression when compared to the control group. Considering the NF- κ B and Nrf2 pathways, our results indicated that treatment with GEN, Ps-NPs or their combination impaired their expression in kidney which significantly upregulated mRNA level of NF- κ B and contrary downregulated mRNA Nrf2 when compared to control group (Fig. 5A–D).

Effects of Ps-NPs or/and GEN exposure on gene expression related to apoptosis

Figure 5 illustrates the changes in kidney gene expression linked to the apoptotic pathway in response to Ps-NPs and/or GEN administration to adult rats. The mRNA level of

Casp-9 was markedly increased in GEN, PS-NPs and their combination compared to controls (Fig. 5G). In contrast, Bax mRNA expression was notably elevated only in rats co-exposed to Ps-NPs and GEN compared to control group (Fig. 5F). On the other hand, as depicted in Fig. 5E, the individual administration of Ps-NPs and GEN as well as their combination resulted in a significant downregulation of Bcl-2 in comparison to control group. Consequently, the ratio of Bax/Bcl-2 was significantly higher in the GEN, Ps-NPs, or their combination compared to the control group, as illustrated in Fig. 5H. This ratio was most significantly pronounced in rats co-exposed to Polystyrene nanoplastics (Ps-NPs).

Histopathological manifestations in the kidney

Histopathological investigation of kidney sections in control and treated groups was elucidated in Figs. 6, 7, 8, 9.

Hematoxylin eosin staining

Hematoxylin and eosin histological stain was used to assess the histological features of the control and treated groups, as depicted in Fig. 6. The control group exhibited normal,

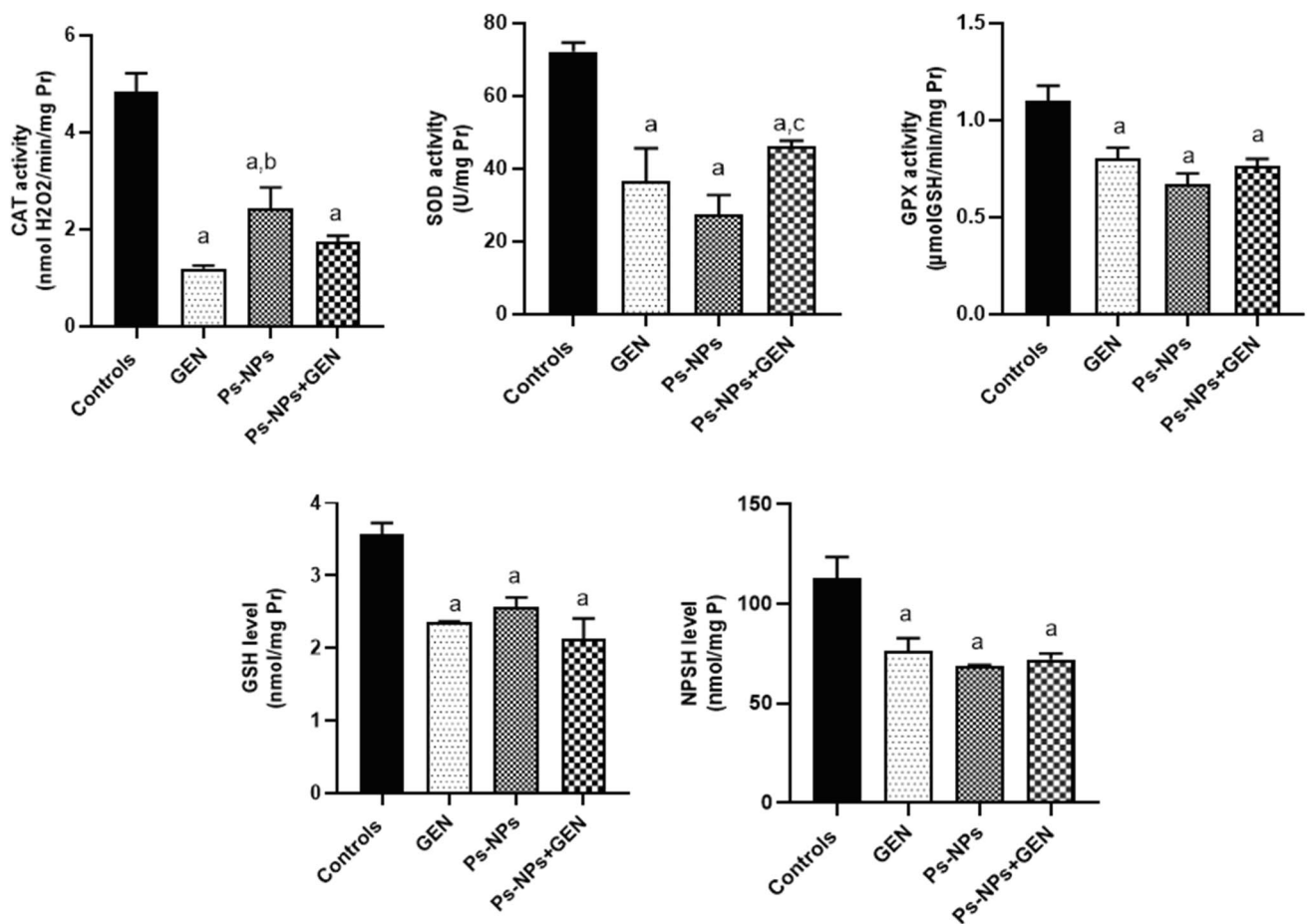


Fig. 4 Effects of Ps-NPs and/or GEN on enzymatic and non-enzymatic antioxidant markers [A—E]. Values are means $\pm$ SD ($n = 10$). **a** $p < 0.05$ compared to controls; **b** $p < 0.05$ compared to GEN group; **c** $p < 0.05$ compared to Ps-NPs group

well-organized kidney tissue histoarchitecture, with intact glomeruli and renal tubules displaying fine, evenly arranged brush borders. In the treated groups (GEN, Ps-NPs or their combination), the kidneys exhibited pathological changes in the renal tubules. Specifically, the kidneys showed signs of tubular injury, including swelling, vacuolization, and necrosis of the tubular epithelial cells. Additionally, there was vascular congestion, irregular lumen, and a minimal of lymphocyte infiltration with the formation of numerous hyaline casts. Cystic dilatation of renal tubules and hyperplastic arteriolo-sclerosis were also observed.

Masson trichrome staining

Histopathological changes in control and treated groups by Masson's trichrome staining are presented in Fig. 7. In the control group, histological structure exhibited normal glomerular and Bowman's capsule, normal renal interstitium

as well as wall of the blood vessels with minimal basophilic collagen fibers surrounding the glomeruli. In contrast, the treated groups (GEN, Ps-NPs and Ps-NPs + GEN) exhibited abundant collagen fibril deposition colorized in green adjacent the renal tubules and encircling the blood vessels.

Sirius-Red staining

Histopathological changes in sections stained by Sirius Red of the kidneys of control, GEN, Ps-NPs or their combination (Fig. 8). The renal tissue of control group shows well-defined glomeruli (G) with a normal distribution of collagen fibers surrounding renal corpuscles and between tubules and normal Bowman's capsule and tubular basement membrane. In contrast, GEN group demonstrated a noticeable expansion of the renal space with a significant deposition of collagen fibers around renal corpuscles and tubules. Tubulointerstitial fibrosis is evident, along with mesangial matrix expansion within the glomeruli (G). The glomeruli in the treated

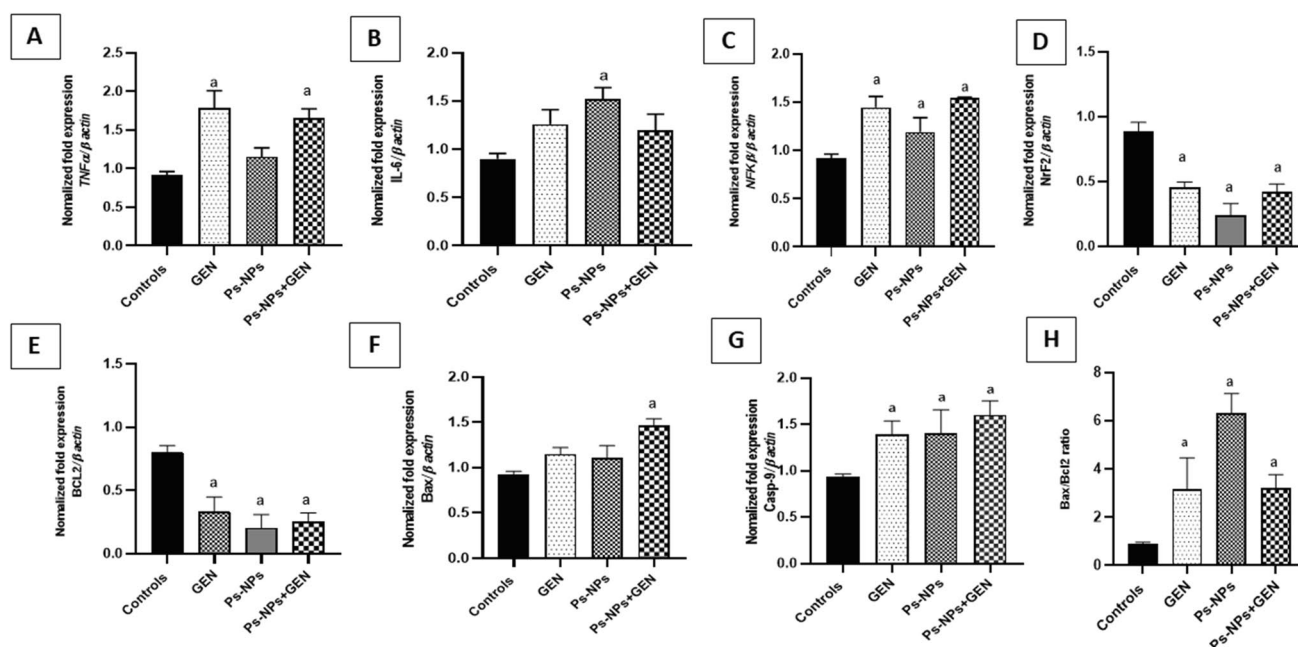


Fig. 5 Gene Expression Level of TNF- α (A), IL-6 (B), Nrf2 (C), NFK β (D), Bcl-2 (E), BAX (F), and Casp-9 (G) measured in the kidney tissues of adult rats exposed to Ps-NPs and/or GEN. The relative gene expression was measured using β -actin as a control. mRNA

expression data are reported as mean $\pm$ SEM ($n=3$). a: $p < 0.05$ compared to controls; b: $p < 0.05$ compared to GEN group; c: $p < 0.05$ compared to Ps-NPs group

groups (Ps-NPs, and Ps-NPs + GEN) were characterized by a thickened glomerular capillary basement membrane, indicating glomerular injury with degenerative changes in Bowman's capsule and a marked tubulointerstitial fibrosis which demonstrating the toxic effects of nanoplastics.

Toluidine blue staining

Histopathological investigation of kidney sections in control and treated rats using toluidine blue are illustrated in Fig. 9. The control group exhibits normal aspects, featuring glomeruli (G), proximal tubules (PCT) that display a narrow lumen with vesicular nuclei (N) containing nucleoli and basal striations. In contrast, distal tubules (DT) are characterized by cuboidal epithelium with rounded vesicular nuclei and basal striations, alongside a normal Bowman's space. Normal podocytes (O) which encase the glomerular capillaries (C), as well as mesangial cells (M) was also observed. In the GEN-treated group, kidney histoarchitecture showed pathological changes with a large renal space and loss of cellular architecture and distorted vesicular nuclei (N). In rats exposed to Ps-NPs and Ps-NPs associated with GEN, degenerative changes and deposits were observed in both PCT and DT. A significant number of tubular cells exhibit signs of degeneration, including discontinuity and vacuolation (V).

Discussion

Polystyrene nanoplastics (Ps-NPs) have become a significant concern in recent years due to their extensive production and diverse applications (Xu et al. 2024; Eliso et al. 2024). These emerging contaminants are now prevalent in the natural environment (Bastante-Rabadán and Boltes 2024). The co-exposure of Ps-NPs with other pollutants, such as antibiotics and heavy metals, can result in synergistic toxicity, which poses a serious threat to human and animal health (Zhang et al. 2024; Zhao et al. 2024; Sun et al. 2024). Polystyrene nanoplastics (Ps-NPs) and antibiotics can accumulate in soil and subsequently enter the food chain, presenting a significant risk to the health of animals and humans (Angon et al. 2024). However, the majority of existing studies have narrowly examined the effects of Ps-NPs and antibiotics such as gentamicin (GEN) on human health when exposure occurs in isolation, without considering the potential synergistic impacts of combined exposure (Ps-NPs + GEN). Further research is essential to investigate the interactions between polystyrene nanoplastics (Ps-NPs) and gentamicin (GEN). Therefore, to better understand the toxic effects and their interaction on kidneys function, we developed rat models exposed to relevant concentrations of Ps-NPs and GEN either individually or in combination.

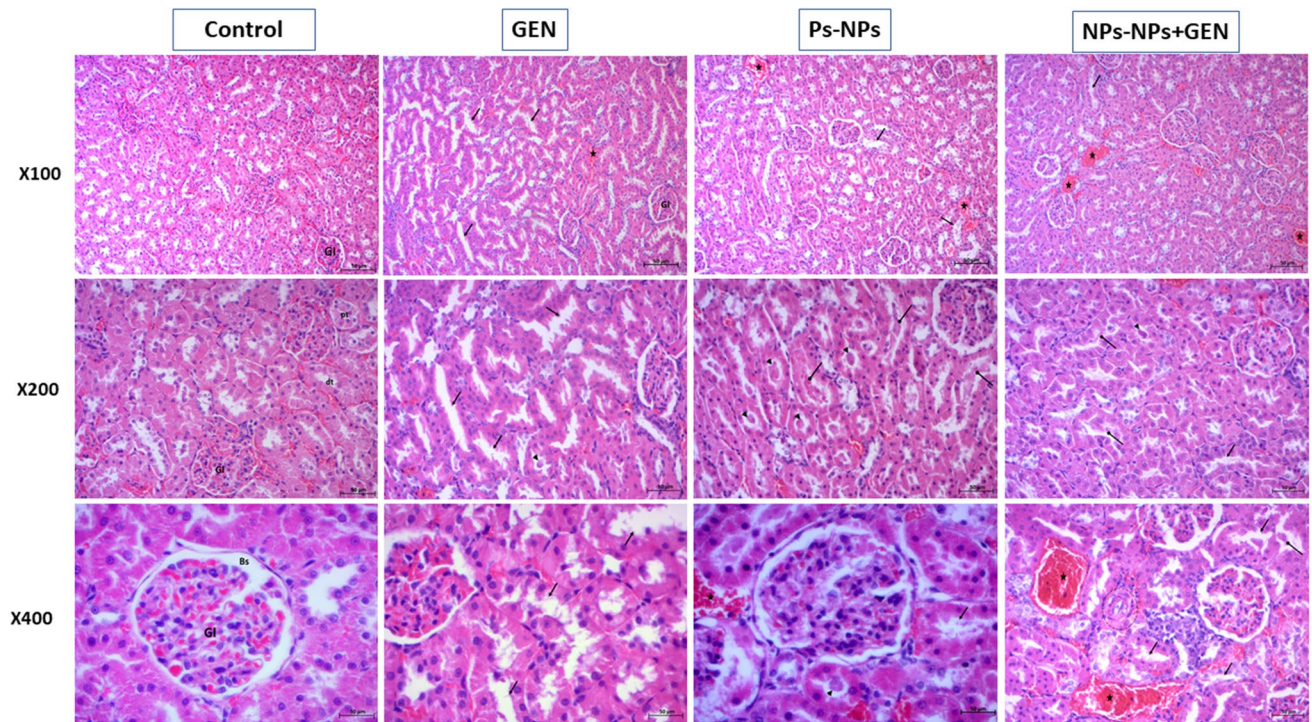


Fig. 6 Representative photographs of kidney sections rats stained with hematoxylin-eosin (Bar=50 μ m). Control group showed normal aspects of renal tubules, proximal (PT) and distal (DT) with intact glomeruli (GI) and Bowman's capsule (bs). Rats treated with GEN and Ps-NPs showed degenerative and dilatation of many renal tubules (black arrow), cast formation (arrowheads), and congestion of RBCs (asterisks). Rats treated with Ps-NPs+GEN showed necrotic cells and denuded tubules (arrow), a lot of congestion of RBCs (asterisks), deposit cellular casts in the enlarged

renal tubules (arrowhead) and hyperplastic arteriolo-sclerosis (circle disconnected); (H&E, Scale bar $\times$ 50 μ m, $\times$ 100, $\times$ 200 and $\times$ 400). GI: Glomérule; PT: Proximal tubule; DT: distal tubule; bs: Bowman's capsule $\rightarrow$ Degenerative renal tubule; $\blacktriangleright$ cast formation; $\star$ Congestion of RBC; $\bullet$ Necrotic cells and denuded tubule; $\circ$ Hyperplastic arteriolo-sclerosis

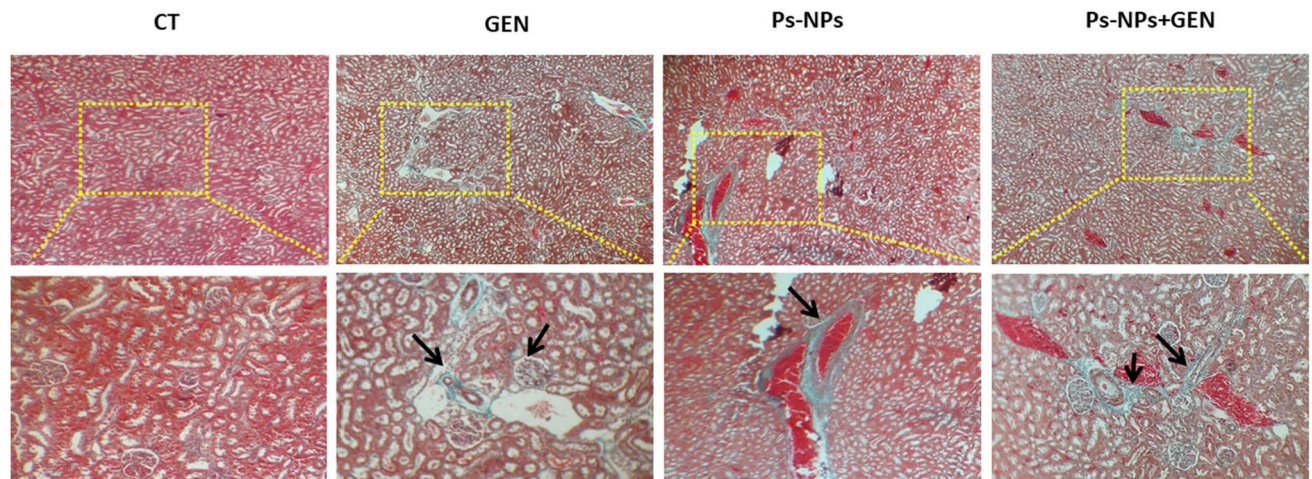


Fig. 7 Representative photographs of Masson's trichrome staining in renal sections of controls and treated groups (GEN, Ps-NPs and Ps-NPs+GEN) (Bar=20 μ m). The control group represented a less collagen deposition indicated by green color in the renal interstitium. While the group treated with GEN and Ps-NPs demonstrated that kidney

tissue was significantly damaged, as evidenced by a marked collagen deposition in renal interstitium (green color) and surrounding fragmented glomeruli and blood vessels; Masson's trichrome, scale bar $\times$ 50 μ m $\times$ 400

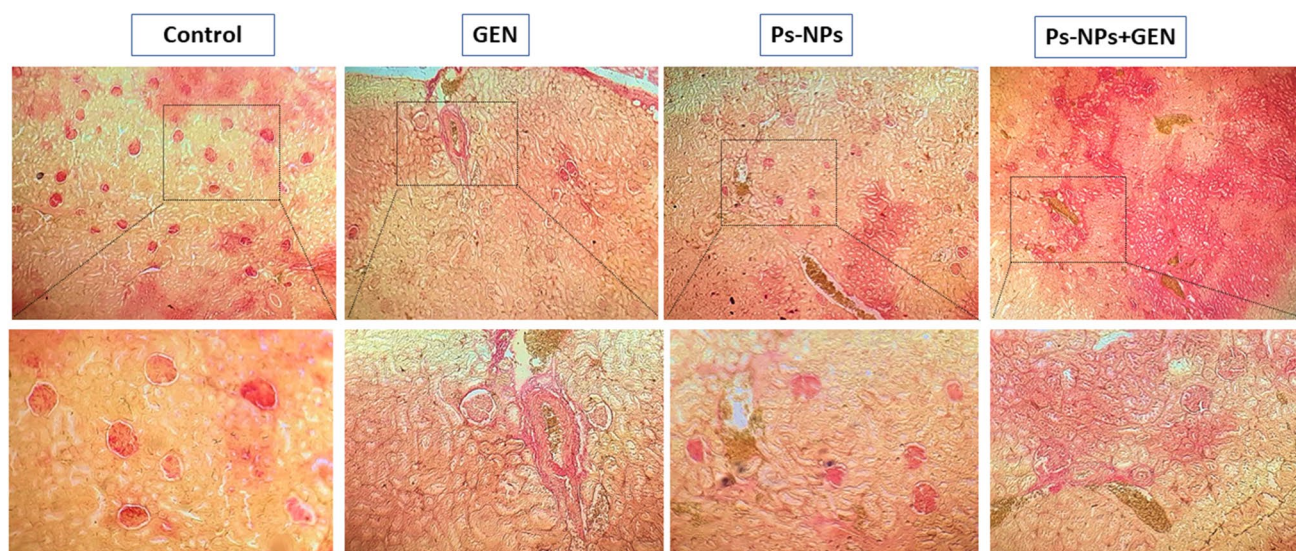


Fig. 8 Representative photographs of kidney histological sections stained with Picro-Sirius red (Bar=20 μ m) of adult rat; control and rats treated with GEN, Ps-NPs and Ps-NPs+GEN. In control group, a normal kidney architecture was observed: well-defined glomeruli (G) with a normal distribution of collagen fibers surrounding renal corpuscles and between tubules (black arrows). The Bowman's capsule and tubular basement membrane appear normal (arrowhead) as well as renal interstitium (asterisks). GEN Group demonstrating a noticeable expansion of the renal space with a significant deposi-

tion of collagen fibers around renal corpuscles and tubules (black arrows). Tubulo-interstitial fibrosis is evident (asterisks), along with mesangial matrix expansion within the glomeruli (G). Ps-NPs, and Ps-NPs+GEN groups are characterized by a thickening of glomerular capillary basement membrane (black arrows), a lesion in the glomeruli with degenerative changes in Bowman's capsule (arrowhead) and a marked tubule-interstitial fibrosis (asterisks); Sirius Red Stain, (Scale bar $\times$ 20 μ m, $\times$ 400 μ m)

In our study, a significant decrease in renal clearance was observed after exposure to Ps-NPs and/or GEN. This effect was accompanied by high levels of plasma creatinine and urea, along with marked reduction in the urinary excretion of these renal function biomarkers. Our results are in agreement with previous studies, which have reported nephrotoxicity induced by either GEN or Ps-NPs, when administered individually to animal model (Ahmed et al. 2022; Wan et al. 2023; Yahaya et al. 2024; Kang et al. 2024).

Based on previous studies reporting the individual toxicity mechanisms of Ps-NPs and GEN, we speculated that oxidative stress, inflammation and apoptosis interplay in the development and progression of nephrotoxic effects caused by the combined exposure of Ps-NPs and GEN. Our findings showed that co-exposure of adult rats to Ps-NPs and GEN enhanced lipid peroxidation, as evidenced by a significant elevation of MDA content, increased protein oxidation level (PCO) as well as an excessive production of hydrogen peroxides (H_2O_2).

To protect itself against oxidative damage resulting from the accumulation of reactive oxygen species (ROS), the kidney has developed a complex and effective system of endogenous antioxidant defenses. The process of mitigating the harmful effects of free radicals primarily involves the antioxidant enzymes including catalase (CAT), superoxide

dismutase (SOD), and glutathione peroxidase (GPx) which they work synergistically to neutralize ROS and, therefore, prevent cellular damage. Our findings indicating an impairment in the oxidant/antioxidant balance of the kidney cells, as evidenced by a significant decrease in CAT, SOD and GPx activities after administration of Ps-NPs, GEN or their combination.

Ps-NPs and/or GEN treatment not only disrupt intracellular antioxidant enzymes but also led to a downregulation of Nrf2 mRNA expression in the kidney of adult rats compared to controls. It is well established that Nrf2 is a transcription factor that accumulates in the nucleus in oxidative and electrophilic stress conditions. It acts as a central regulator for the expression of several enzymatic and non-enzymatic antioxidant genes, which are essential for enhancing cell survival and facilitating recovery from oxidative stress (Manna et al. 2023).

Given that the glutathione is considered one of the primary line defense against oxidative damage (Labarrere and Kassab 2022), we hypothesized that oxidative stress induced by Ps-NPs and/or GEN may disrupt the homeostasis of total thiol groups (t-SH). In our study, exposure of rats to Ps-NPs and/or GEN was marked by a significant decrease in GSH and NPSH levels. These results clearly exhibited that impairment in the pro-oxidant/antioxidant balance and glutathione homeostasis

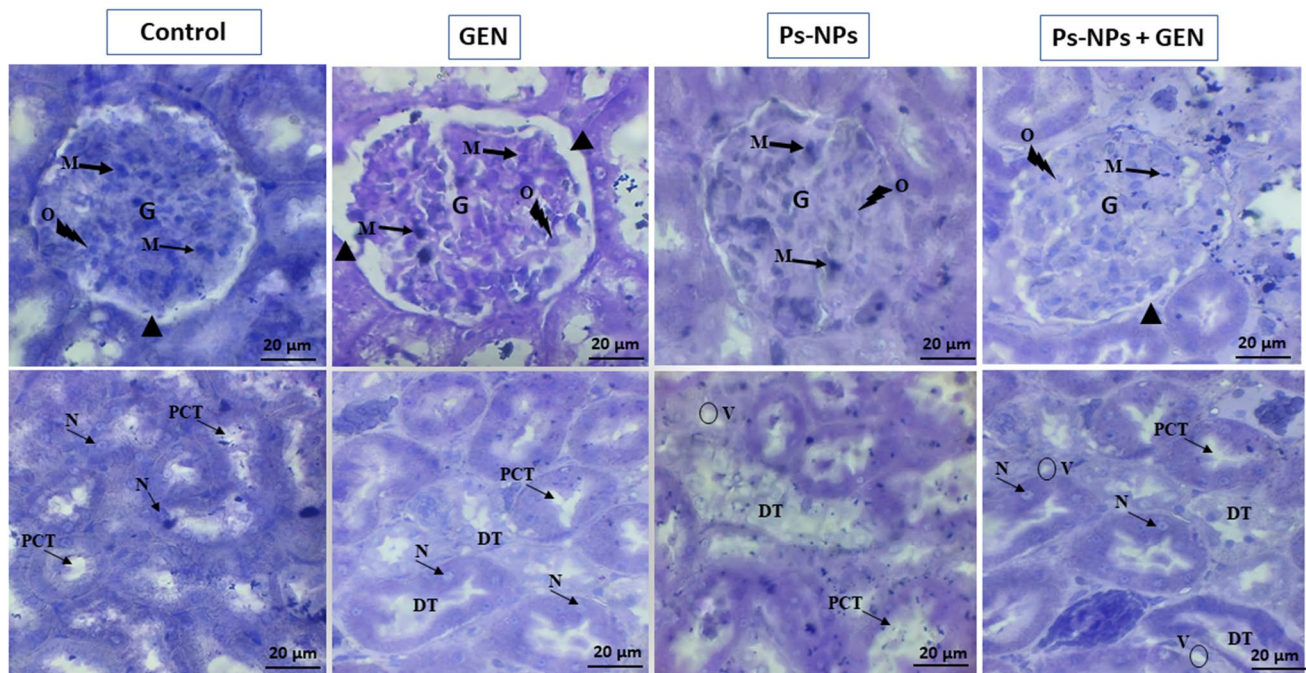


Fig. 9 Representative photographs of toluidine blue stained sections of rat kidney of different groups: control and treated groups (GEN, Ps-NPs, and Ps-NPs + GEN) (Bar = 20 µm). The control group exhibits a normal histological aspect, featuring glomeruli (G), proximal (PCT) and distal tubules (DT), podocytes (O) lining the visceral layer and covering glomerular capillaries (C), as well as mesangial cells (M). PCT displays a narrow lumen with pyramidal cells containing vesicular nuclei (N) with basal striations. In contrast, DT are characterized by cuboidal epithelium lining at their borders with central

nuclei and basal striations, alongside a normal Bowman's space indicated by arrowheads. In GEN-treated group, the glomeruli exhibit an enlarged Bowman's space (arrowhead) and loss of cellular architecture and distorted vesicular nuclei (N). In Ps-NPs and Ps-NPs + GEN groups, degenerative changes and deposits were evident in PCT and DT, a significant number of tubular cells exhibit signs of degeneration, including discontinuity of cell membranes and vacuolation (V); (Toluidine Blue, Scale bar $\times 20$ µm, $\times 400$)

are important mechanisms of GEN and Ps-NPs co-exposure-induced nephrotoxicity. Our results are basically aligned with several studies which demonstrated the harmful effect of Ps-NPs (Hwang et al. 2022; Yang et al. 2024) and GEN (Kamei et al. 2015; Bolten et al. 2023) on the kidney of experimental models.

It is well documented that oxidative stress-driven inflammatory processes lead to several pathologic conditions, such as cardiovascular diseases, cancer, neurodegenerative disease (Bahari et al. 2023) and renal dysfunction (Elgendy et al. 2024). One possible mechanism for heightened inflammation is the activation of NF- κ B during ROS production and kidney damage, which underscores inflammation-related oxidative stress in the progression of disease. The transcription factor NF- κ B is essential for activating downstream genes that produce pro-inflammatory cytokines, thereby mediating various inflammatory responses and pathways that lead to cell death. Tumor necrosis factor- α (TNF- α), a key pro-inflammatory cytokine, can trigger fibroblasts, endothelial cells, and macrophages to release chemokines, contributing to tissue damage and the persistence of chronic inflammation (Adinew et al. 2023). In our experimental study, exposure

of rats to Ps-NPs, GEN or their combination resulted in the expression of inflammatory cytokines in kidney. Compared to the control group, our study showed that exposure to GEN and both PS-NPs and GEN induced an increase of the mRNA levels of TNF- α , interleukin-6 (IL-6) and NF- κ B, while exposure to PS-NPs alone has no effect on the expression of inflammatory factors. These results indicate that Ps-NPs significantly exacerbate GEN-induced inflammation in kidney tissue. Our findings were in agreement with those reported by Fu et al. (2023) showing that combined exposure to nanoplastics and other pollutants induced NF- κ B activation triggering in turn a cascade of pro-inflammatory transcription such as TNF- α . In addition, our findings further demonstrated that exposure to both Ps-NPs and GEN significantly increased collagen renal fibers deposition indicating that co-exposure to these pollutants enhanced tissue structure damages and fibrosis.

Oxidative stress and inflammation trigger apoptosis in aquatic organisms (Pei et al. 2024) and in testicular rats (Tuncer et al. 2023). Apoptosis is a controlled process of programmed cell death, which is required to maintain various physiological functions. Among the apoptotic indicators,

Bcl-2 family proteins and caspase enzymes play a crucial role in the process of cellular death (Kaloni et al. 2023). During apoptosis, the pro-apoptotic protein Bax, when activated, translocates to the exterior mitochondrial membrane. As a result, the mitochondria membrane's integrity changed and became more permeable, leading to the leakage of cytochrome c (Cyt c) into the cytoplasm, triggering consequently a cascade of caspases and accelerating apoptosis. Additionally, the tumor suppressor protein p53 can promote the expression of apoptosis-related genes including Bax and Bcl-2. Interestingly, our results showed that there were no significant changes in the expression of Bax and Caspase-9 after Ps-NPs and GEN when administered separately to adult rats; however, they increased prominently in the combined group (Ps-NPs + GEN). The impairment of gene expression related to apoptosis is directly linked to mitochondrial dysfunction (Li et al. 2023). Our findings suggest that, in contrast to their individual effects, GEN and Ps-NPs may interact together leading to heightened toxicity, as an additive response. Similar results were observed in previous *in vitro* studies, which have reported that polystyrene nanoplastics promote the apoptosis in CaCo-2 and HepG2 cells induced by triphenyl phosphate (TPHP) and Okadaic (Yan et al. 2023; Huang et al. 2024). Our results align with previous studies conducted on liver rat and mouse kidney cells after either Polystyrene nanoparticles (Yasin et al. 2022) or arsenic and polystyrene-nanoplastics (Zhong et al. 2022) treatments, respectively.

As expected, the biochemical and inflammation/apoptosis findings are correlated with renal structural changes and leading ultimately to dilatation of renal tubules and kidney infiltration of inflammatory leukocytes localized around the blood vessels in GEN and Ps-NPs treated rats. Marked deposit cellular casts were observed in the enlarged renal tubules, along with hyperplastic arteriolo-sclerosis as evidenced by a marked collagen deposition in renal interstitium and glomeruli fragmentation disorder were shown in both GEN and Ps-NPs treated rats. Our present study confirmed the previous findings of Meng et al. (2023) and Tang et al. (2023) who have shown renal structural alterations in the kidney of mice following polystyrene nanoplastics treatment.

Conclusions

Our present study provides new insights into the possible renal effects of co-exposure to gentamicin (GEN) and polystyrene nanoparticles (PS-NPs). This paper is the first to demonstrate interactions between GEN and PS-NPS. Accordingly, it suggests that the combination of GEN and PS-NPS may lead to underlying mechanisms affecting kidney function, as evidenced by inflammation mediated by pro-inflammatory cytokines such as TNF- α and IL-6, activation of NF- κ B signaling, and modulation of Nrf2. These inflammatory responses

can exacerbate oxidative stress within renal tissues, contributing to further cellular damage. Additionally, our findings indicate that co-exposure results in mitochondria-mediated apoptosis, characterized by the overexpression of Bax and caspase-9. Histopathological analysis revealed significant structural disorders in the kidneys, further supporting the harmful effect of combined GEN and PS-NPs exposure on renal function. Our findings offer new insights for future mechanistic and clinical studies on the risks associated with concurrent environmental co-exposure to Ps-NPs and residues of pharmaceutical products in Human.

Acknowledgements This work was supported by the Tunisian Ministry of Higher Education and Scientific Research.

Author contribution Khadija Boukholda : Writing and editing original draft, Methodology, Conceptualization; Awatef Elwej : Methodology, Investigation, Conceptualization; Sabrina Ben Slimen : contribute to the oxidative stress evaluations and data analysis; Abir Mhadhbi : Methodology, Investigation, Conceptualization, Rim Marrekchi : analysis of biochemical parameters; Ons Boudawara : interpreted histological results ; Bülent Kaya : review and edit-ing; Michèle Bouchard: review and edit-ing ; Hamadi Fetoui : Methodology, Writing and editing manuscript, Conceptualization. The authors declare that all data were generated in-house and that no paper mill was used.

Data availability All source data for this work (or generated in this study) are available upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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